Lab06-Composite

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VNU-HCM, HCMUS, FIT, SE

CS202 - Programming Systems

25A01 - 02

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A: YY: 01

H: YY: 03

> 💡 Students are permitted to use AI assistance for this series of homework.
> If you do not use AI: Please include Declaration A in your README.md file: "I did not use AI for this homework."
> If you use AI: Please include Declaration B in your README.md file: "I did use AI for this homework." The expected workflow is that you prompt the AI, it generates a plan, you review that plan, the AI executes the action, and you review, edit, and re-prompt as needed. Ultimately, you bear full responsibility for the final result.

Design class diagrams using Mermaid format and implement the required functionality in C++ for the following assignments. For your final submission, please ensure you include the C++ source code, the raw Mermaid syntax, and the rendered image files for each class diagram.

Class diagram: https://en.wikipedia.org/wiki/Class_diagram

Use stereotypes to annotate the meaning of classes and methods.

```
<<interface>>, <<abstract>>, <<concrete>>, <<constructor>>,
<<destructor>>, <<friend>>, <<static>>, <<virtual>>, <<pure
virtual>>, <<override>>, <<operator>>.
```

```
Ex: +<<pure virtual>> getSalary(): double
```

Assignment 1 – LMS

You are asked to develop a system for managing the learning content offered in its online courses. Each course may contain different types of learning materials, such as instructional videos and quizzes. These learning materials share several common characteristics, including a **title**, an estimated completion **duration**, and the ability to display their information. However, each type of learning material also has its own specific attributes and rules.

- A **Video** represents a recorded instructional lesson. In addition to its title, each video has a fixed duration measured in minutes. When its information is displayed, the system should show the title of the video and its duration.
- A **Quiz** represents an assessment containing a specified number of questions. The estimated completion duration of a quiz is calculated based on the number of questions, assuming that a student requires approximately two minutes to complete each question. When its information is displayed, the system should show the title of the quiz, the number of questions, and its estimated completion duration.
- The company also organizes its learning content into **Modules**. A module has a title and may contain multiple learning items. These items may be videos, quizzes, or even other modules. Therefore, the system must support modules nested inside other modules. The total duration of a module is calculated as the sum of the durations of all learning items contained directly or indirectly within it.

For example, a module titled "**Introduction to Object-Oriented Programming**" may contain:

- a video titled "**Classes and Objects**" with a duration of **25 minutes**;
- a quiz titled "**Basic OOP Concepts**" containing **5 questions**;

- another module titled "**Inheritance and Polymorphism**", which contains additional videos and quizzes.

The system must allow videos, quizzes, and modules to be treated uniformly as learning items. A client should be able to request the duration or display the details of any learning item without needing to determine whether the object is a video, a quiz, or a module.

When displaying a module, the program should display the module's **title** and **total duration**, followed by the details of **all its child learning items**. Nested learning items should be **indented** appropriately so that the hierarchical structure of the course content can be clearly seen.

An example output could be (please note the indentation):

```
[Module] Introduction to Object-Oriented Programming (1h 15
m)
  [Video] Classes and Objects (0h 25m)
  [Quiz] Basic OOP Concepts (5 questions, 0h 10m)
  [Module] Inheritance and Polymorphism (0h 40m)
    [Video] Introduction to Inheritance (0h 20m)
    [Quiz] Polymorphism Quiz (10 questions, 0h 20m)
```

The design for the system should be flexible and ready for future expansion of additional types of learning content, such as programming exercises, reading materials, assignments, or interactive labs without significantly modifying the existing program. The system must also apply the inheritance and polymorphism. It should also apply the design pattern so that individual learning items and groups of learning items can be processed through the same common interface.

The program must properly manage dynamically created objects. The ownership and lifetime of learning items should be clearly defined, and all allocated memory must be released correctly to prevent memory leaks, dangling pointers, double deletion, or application crashes.

You are asked to complete the following tasks:

1. Load a list of learning items from a text-based file. Provide your input file in the submission
2. Displays the complete hierarchical structure of a module with suitable indentation using runtime polymorphism with detailed information for each

learning item including their titles, durations or number of questions if it is a quiz.

Notes:

- A quiz requires two minutes per question.
- A module's duration is the sum of the durations of all its children.
- A module may contain videos, quizzes, or other modules.

The implementation must use runtime polymorphism rather than explicit type checking.

Assignment 2 – Advanced Features in Python

Write a presentation (Markdown format) with at least 20 slides to explain the advanced OOP concepts in Python such as:

- Multi-inheritance
- Interface
- Inner class
- Nested class
- Lamda expression

Assignment 3 – Advanced Features in Java

Write a presentation (Markdown format) with at least 20 slides to explain the advanced OOP concepts in Java such as:

- Multi-inheritance
- Interface
- Inner class
- Nested class
- Lamda expression